QSAR-X

Detailed workflow, training output, validation output, and target-wise model report

Purpose

This is the detailed companion report. It records every aggregate numeric output generated for each currently registered target-specific model: data volume, scaffold counts, train/validation/test split sizes, cross-validation comparison means and standard deviations, selected model, validation metrics, scaffold-held-out metrics, external-source metrics, structure-overlap removal count, and every external quality-gate result. It does not include all molecule-level assay rows or prediction rows because those are datasets rather than aggregate outputs; they remain in the local CSV artifacts and can be exported through the application.

Truthful deployment status: 20 target-specific models are loaded for screening. No model has passed every strict external validation gate. The detailed tables must not be interpreted as proof of commercial or prospective experimental accuracy.

Contents

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1. Pipeline and data lineage

Input: target choice plus molecular SMILES. For each target, ChEMBL target metadata and IC50 assay activities were retrieved. The curation path retained direct single-protein IC50 values, exact relation, positive numeric values, and nM units; structures were canonicalized using RDKit and duplicate structures were aggregated. pIC50 is calculated as 9 - log10(IC50 nM).

Training: chirality-aware Bemis-Murcko scaffold groups were split approximately 64/16/20 for training/validation/scaffold test. Training CV compared MedianBaseline, ExtraTrees, RandomForest, and SVR. The lowest train-CV RMSE candidate was saved as candidate_model.joblib. Current selected candidate family is ExtraTrees for the registered models.

Validation: BindingDB REST was used for an external-source non-overlap check by UniProt accession. Exact structures found in ChEMBL training were removed. PubChem BioAssay discovery and strict ingestion were also implemented; only non-ChEMBL sources with an explicit numeric IC50, unit, exact relation, SMILES, assay context, and publication provenance can be considered for the strict validator.

Software stack

Data and chemistry	ChEMBL activity endpoint data; BindingDB REST external-source checks; PubChem BioAssay PUG-REST discovery; RDKit canonicalization and Morgan fingerprints
Training	Python 3.12.14; scikit-learn 1.9.1; RDKit 2026.09.1; ExtraTrees / RandomForest / SVR comparison
Service	FastAPI versioned target registry and React/Vite web interface

2. Aggregate target inventory

All values below come from the currently registered model version for each target. External n/RMSE are from the latest external report. “Not passed” means at least one strict quality gate is false.

ABL1	CHEMBL1862	baseline_v1	2151	0.683	139	1.146	0.934	0.556	Not passed
AKT1	CHEMBL4282	baseline_v1	3455	0.731	91	0.792	0.517	0.608	Not passed
BRAF	CHEMBL5145	baseline_v2	5384	0.656	203	1.228	1.010	0.434	Not passed
CDK2	CHEMBL301	baseline_v1	2115	0.843	741	1.040	0.800	0.489	Not passed
EGFR	CHEMBL203	baseline_v1	10834	0.850	770	1.665	1.390	0.366	Not passed
ERBB2	CHEMBL1824	baseline_v1	2522	0.741	1497	1.404	1.087	0.215	Not passed
FGFR1	CHEMBL3650	baseline_v1	3596	0.657	103	0.749	0.611	0.534	Not passed
FLT3	CHEMBL1974	baseline_v1	3766	0.782	52	0.861	0.596	0.723	Not passed
IGF1R	CHEMBL1957	baseline_v1	2324	0.593	33	0.727	0.565	0.461	Not passed
JAK2	CHEMBL2971	baseline_v1	10405	0.696	10	0.333	0.243	0.673	Not passed
KDR	CHEMBL279	baseline_v1	9762	0.759	160	0.829	0.650	0.653	Not passed
KIT	CHEMBL1936	baseline_v1	1946	0.750	128	1.010	0.798	-0.175	Not passed
MAP2K1	CHEMBL3587	baseline_v1	866	0.767	63	0.751	0.627	0.606	Not passed
MAPK14	CHEMBL260	baseline_v1	4744	0.720	245	1.148	0.858	0.435	Not passed
MET	CHEMBL3717	baseline_v1	4193	0.647	83	0.813	0.494	0.820	Not passed
MTOR	CHEMBL2842	baseline_v1	4354	0.603	28	0.707	0.582	0.722	Not passed
PDGFRA	CHEMBL2007	baseline_v1	632	0.798	71	1.073	0.884	-0.084	Not passed
PIK3CA	CHEMBL4005	baseline_v1	6465	0.616	566	1.202	0.958	0.396	Not passed
RET	CHEMBL2041	baseline_v1	3980	0.612	95	1.413	1.151	0.278	Not passed
SRC	CHEMBL267	baseline_v1	3601	0.841	62	0.638	0.485	0.888	Not passed

3.1. ABL1 - CHEMBL1862

Protein: Tyrosine-protein kinase ABL1 | **UniProt:** P00519 | **Model version:** baseline_v1 | **Selected model:** SVR

Dataset and split output

Curated compound count	2151
Scaffold count	932
Train / validation / scaffold test	1406 / 343 / 402
Train / validation / test scaffolds	596 / 149 / 187
Fingerprint	Morgan, radius 2, 2,048 bits
Candidate model SHA256	cc323f351eaa3554561e759360cdd30919099b60c86b4a4bdee01558eb1333c7

Cross-validation comparison output

MedianBaseline	-0.050	0.076	1.494	0.049	1.272	0.036
ExtraTrees	0.671	0.026	0.837	0.044	0.617	0.038
RandomForest	0.669	0.030	0.838	0.047	0.621	0.037
SVR	0.684	0.030	0.819	0.042	0.619	0.036

Selected candidate internal output

Validation	0.749	0.752	0.564
Scaffold-held-out test	0.798	0.683	0.512

External-source validation output

Source	BindingDB REST
Exact training-structure overlaps removed	253
External non-overlap n	139
External R2 / RMSE / MAE / Spearman	0.335 / 1.146 / 0.934 / 0.556
accession matches	True
exact structure overlap removed	True
minimum 30 external compounds	True
relation and assay context documented	False
Overall strict quality-gates passed	False
External status	external_source_exploratory_only

3.2. AKT1 - CHEMBL4282

Protein: RAC-alpha serine/threonine-protein kinase | **UniProt:** P31749 | **Model version:** baseline_v1 | **Selected model:** SVR

Dataset and split output

Curated compound count	3455
Scaffold count	1284
Train / validation / scaffold test	2207 / 629 / 619
Train / validation / test scaffolds	821 / 206 / 257
Fingerprint	Morgan, radius 2, 2,048 bits
Candidate model SHA256	50306f5d69a3ce2dcb31e82f7ec5fb2fae721f23bab92f5e35652f43c0f6512a

Cross-validation comparison output

MedianBaseline	-0.007	0.010	1.201	0.007	0.960	0.028
ExtraTrees	0.529	0.061	0.820	0.057	0.620	0.038
RandomForest	0.531	0.043	0.819	0.041	0.626	0.026
SVR	0.568	0.051	0.786	0.051	0.604	0.038

Selected candidate internal output

Validation	0.723	0.728	0.558
Scaffold-held-out test	0.668	0.731	0.566

External-source validation output

Source	BindingDB REST
Exact training-structure overlaps removed	406
External non-overlap n	91
External R2 / RMSE / MAE / Spearman	0.396 / 0.792 / 0.517 / 0.608
accession matches	True
exact structure overlap removed	True
minimum 30 external compounds	True
relation and assay context documented	False
Overall strict quality-gates passed	False
External status	external_source_exploratory_only

3.3. BRAF - CHEMBL5145

Protein: Serine/threonine-protein kinase B-raf | **UniProt:** P15056 | **Model version:** baseline_v2 | **Selected model:** SVR

Dataset and split output

Curated compound count	5384
Scaffold count	1901
Train / validation / scaffold test	3376 / 743 / 1265
Train / validation / test scaffolds	1216 / 304 / 381
Fingerprint	Morgan, radius 2, 2,048 bits
Candidate model SHA256	3ded4d9f61ba03b6682c7dc0538b64e327c5abf81f2a0072dfddf5d5aba5af3f

Cross-validation comparison output

MedianBaseline	-0.052	0.063	1.287	0.029	1.038	0.012
ExtraTrees	0.598	0.078	0.794	0.085	0.583	0.084
RandomForest	0.592	0.067	0.801	0.074	0.590	0.077
SVR	0.602	0.056	0.791	0.064	0.578	0.069

Selected candidate internal output

Validation	0.725	0.626	0.479
Scaffold-held-out test	0.720	0.656	0.483

External-source validation output

Source	BindingDB REST
Exact training-structure overlaps removed	682
External non-overlap n	203
External R2 / RMSE / MAE / Spearman	0.144 / 1.228 / 1.010 / 0.434
accession matches	True
exact structure overlap removed	True
minimum 30 external compounds	True
relation and assay context documented	False
Overall strict quality-gates passed	False
External status	external_source_exploratory_only

3.4. CDK2 - CHEMBL301

Protein: Cyclin-dependent kinase 2 | **UniProt:** P24941 | **Model version:** baseline_v1 | **Selected model:** SVR

Dataset and split output

Curated compound count	2115
Scaffold count	921
Train / validation / scaffold test	1182 / 318 / 615
Train / validation / test scaffolds	588 / 148 / 185
Fingerprint	Morgan, radius 2, 2,048 bits
Candidate model SHA256	fe4bd91b06ca88966bd1c57de38f041b5ac058b782ad0f21743c4f191a3d8110

Cross-validation comparison output

MedianBaseline	-0.078	0.080	1.299	0.124	1.060	0.072
ExtraTrees	0.376	0.017	0.987	0.061	0.746	0.030
RandomForest	0.364	0.049	0.997	0.094	0.775	0.059
SVR	0.433	0.036	0.941	0.077	0.731	0.050

Selected candidate internal output

Validation	0.670	0.756	0.594
Scaffold-held-out test	0.569	0.843	0.638

External-source validation output

Source	BindingDB REST
Exact training-structure overlaps removed	292
External non-overlap n	741
External R2 / RMSE / MAE / Spearman	0.199 / 1.040 / 0.800 / 0.489
accession matches	True
exact structure overlap removed	True
minimum 30 external compounds	True
relation and assay context documented	False
Overall strict quality-gates passed	False
External status	external_source_exploratory_only

3.5. EGFR - CHEMBL203

Protein: Epidermal growth factor receptor | **UniProt:** P00533 | **Model version:** baseline_v1 | **Selected model:** ExtraTrees

Dataset and split output

Curated compound count	10834
Scaffold count	3977
Train / validation / scaffold test	7023 / 1443 / 2368
Train / validation / test scaffolds	2544 / 637 / 796
Fingerprint	Morgan, radius 2, 2,048 bits
Candidate model SHA256	c91b2692dbabdc28db64972701832948ba5d4dd4902bc5bcdb45e638344a363f

Cross-validation comparison output

MedianBaseline	-0.008	0.006	1.380	0.061	1.143	0.054
ExtraTrees	0.569	0.016	0.902	0.051	0.681	0.022
RandomForest	0.561	0.018	0.911	0.052	0.693	0.022
SVR	0.563	0.030	0.907	0.016	0.689	0.011

Selected candidate internal output

Validation	0.599	0.851	0.613
Scaffold-held-out test	0.552	0.850	0.635

External-source validation output

Source	BindingDB REST
Exact training-structure overlaps removed	1414
External non-overlap n	770
External R2 / RMSE / MAE / Spearman	-0.386 / 1.665 / 1.390 / 0.366
accession matches	True
exact structure overlap removed	True
minimum 30 external compounds	True
relation and assay context documented	False
Overall strict quality-gates passed	False
External status	external_source_exploratory_only

3.6. ERBB2 - CHEMBL1824

Protein: Receptor tyrosine-protein kinase erbB-2 | **UniProt:** P04626 | **Model version:** baseline_v1 | **Selected model:** ExtraTrees

Dataset and split output

Curated compound count	2522
Scaffold count	1063
Train / validation / scaffold test	1537 / 511 / 474
Train / validation / test scaffolds	680 / 170 / 213
Fingerprint	Morgan, radius 2, 2,048 bits
Candidate model SHA256	4d1b7dadd41eba0818d7d86e33448e6a423d8fd35c52bf99e7118081a6003d69

Cross-validation comparison output

MedianBaseline	-0.018	0.011	1.132	0.072	0.926	0.077
ExtraTrees	0.553	0.063	0.747	0.016	0.563	0.006
RandomForest	0.552	0.064	0.748	0.021	0.570	0.004
SVR	0.529	0.068	0.767	0.022	0.588	0.003

Selected candidate internal output

Validation	0.690	0.685	0.532
Scaffold-held-out test	0.629	0.741	0.571

External-source validation output

Source	BindingDB REST
Exact training-structure overlaps removed	461
External non-overlap n	1497
External R2 / RMSE / MAE / Spearman	-0.280 / 1.404 / 1.087 / 0.215
accession matches	True
exact structure overlap removed	True
minimum 30 external compounds	True
relation and assay context documented	False
Overall strict quality-gates passed	False
External status	external_source_exploratory_only

3.7. FGFR1 - CHEMBL3650

Protein: Fibroblast growth factor receptor 1 | **UniProt:** P11362 | **Model version:** baseline_v1 | **Selected model:** SVR

Dataset and split output

Curated compound count	3596
Scaffold count	1380
Train / validation / scaffold test	2398 / 588 / 610
Train / validation / test scaffolds	883 / 221 / 276
Fingerprint	Morgan, radius 2, 2,048 bits
Candidate model SHA256	97cdd5d3357b934f64abf2bb1a27bbc91453f456b99431db55dda15a7f84d561

Cross-validation comparison output

MedianBaseline	-0.028	0.020	1.192	0.152	0.959	0.133
ExtraTrees	0.591	0.076	0.743	0.014	0.571	0.022
RandomForest	0.567	0.071	0.766	0.031	0.591	0.035
SVR	0.629	0.053	0.710	0.037	0.559	0.033

Selected candidate internal output

Validation	0.645	0.647	0.475
Scaffold-held-out test	0.695	0.657	0.505

External-source validation output

Source	BindingDB REST
Exact training-structure overlaps removed	369
External non-overlap n	103
External R2 / RMSE / MAE / Spearman	0.389 / 0.749 / 0.611 / 0.534
accession matches	True
exact structure overlap removed	True
minimum 30 external compounds	True
relation and assay context documented	False
Overall strict quality-gates passed	False
External status	external_source_exploratory_only

3.8. FLT3 - CHEMBL1974

Protein: Receptor-type tyrosine-protein kinase FLT3 | **UniProt:** P36888 | **Model version:** baseline_v1 | **Selected model:** SVR

Dataset and split output

Curated compound count	3766
Scaffold count	1830
Train / validation / scaffold test	2517 / 558 / 691
Train / validation / test scaffolds	1171 / 293 / 366
Fingerprint	Morgan, radius 2, 2,048 bits
Candidate model SHA256	992711837a71e5bd5fc72685139285a3147ca576258ed1f50874185170f10ac5

Cross-validation comparison output

MedianBaseline	-0.036	0.031	1.266	0.073	1.048	0.059
ExtraTrees	0.498	0.104	0.875	0.058	0.669	0.057
RandomForest	0.496	0.091	0.878	0.053	0.676	0.054
SVR	0.531	0.078	0.848	0.052	0.653	0.054

Selected candidate internal output

Validation	0.679	0.720	0.551
Scaffold-held-out test	0.608	0.782	0.606

External-source validation output

Source	BindingDB REST
Exact training-structure overlaps removed	440
External non-overlap n	52
External R2 / RMSE / MAE / Spearman	0.552 / 0.861 / 0.596 / 0.723
accession matches	True
exact structure overlap removed	True
minimum 30 external compounds	True
relation and assay context documented	False
Overall strict quality-gates passed	False
External status	external_source_exploratory_only

3.9. IGF1R - CHEMBL1957

Protein: Insulin-like growth factor 1 receptor | **UniProt:** P08069 | **Model version:** baseline_v1 | **Selected model:** SVR

Dataset and split output

Curated compound count	2324
Scaffold count	905
Train / validation / scaffold test	1548 / 309 / 467
Train / validation / test scaffolds	579 / 145 / 181
Fingerprint	Morgan, radius 2, 2,048 bits
Candidate model SHA256	54fb482c981a0d4a96553f00e842fef1152fe2b17783fc7e23cf1450bb9b311d

Cross-validation comparison output

MedianBaseline	-0.015	0.008	1.262	0.127	1.008	0.128
ExtraTrees	0.729	0.016	0.652	0.068	0.498	0.061
RandomForest	0.732	0.009	0.649	0.071	0.499	0.059
SVR	0.757	0.018	0.616	0.046	0.473	0.034

Selected candidate internal output

Validation	0.602	0.799	0.566
Scaffold-held-out test	0.769	0.593	0.410

External-source validation output

Source	BindingDB REST
Exact training-structure overlaps removed	183
External non-overlap n	33
External R2 / RMSE / MAE / Spearman	0.136 / 0.727 / 0.565 / 0.461
accession matches	True
exact structure overlap removed	True
minimum 30 external compounds	True
relation and assay context documented	False
Overall strict quality-gates passed	False
External status	external_source_exploratory_only

3.10. JAK2 - CHEMBL2971

Protein: Tyrosine-protein kinase JAK2 | **UniProt:** O60674 | **Model version:** baseline_v1 | **Selected model:** ExtraTrees

Dataset and split output

Curated compound count	10405
Scaffold count	3828
Train / validation / scaffold test	7003 / 1385 / 2017
Train / validation / test scaffolds	2449 / 613 / 766
Fingerprint	Morgan, radius 2, 2,048 bits
Candidate model SHA256	f1529fdf9ec74b2de266ee5b1cc636be42fc1ea4dfe7369e706622ec93416806

Cross-validation comparison output

MedianBaseline	-0.010	0.016	1.302	0.077	1.042	0.072
ExtraTrees	0.685	0.036	0.725	0.010	0.531	0.005
RandomForest	0.681	0.036	0.730	0.009	0.540	0.005
SVR	0.684	0.039	0.726	0.008	0.543	0.005

Selected candidate internal output

Validation	0.685	0.716	0.515
Scaffold-held-out test	0.716	0.696	0.513

External-source validation output

Source	PubChem BioAssay AID 1699
Exact training-structure overlaps removed	95
External non-overlap n	10
External R2 / RMSE / MAE / Spearman	0.163 / 0.333 / 0.243 / 0.673
target chembl id matches	True
target accession matches	True
endpoint is ic50	True
units are nm	True
relation exact	True
assay context documented	True
publication id documented	True
no compound overlap with training	True
no publication overlap with training	True

minimum 30 external compounds	False
metrics meet screening floor	False
model version and hash match	True
external source provided	True
Overall strict quality-gates passed	False
External status	external_validation_gates_failed

3.11. KDR - CHEMBL279

Protein: Vascular endothelial growth factor receptor 2 | **UniProt:** P35968 | **Model version:** baseline_v1 | **Selected model:** ExtraTrees

Dataset and split output

Curated compound count	9762
Scaffold count	3663
Train / validation / scaffold test	6588 / 1265 / 1909
Train / validation / test scaffolds	2344 / 586 / 733
Fingerprint	Morgan, radius 2, 2,048 bits
Candidate model SHA256	15d9153c231fbd8ce07a7828045581703cb2d9aeac30ac45ef15f67556e8fc6a

Cross-validation comparison output

MedianBaseline	-0.029	0.029	1.159	0.080	0.943	0.079
ExtraTrees	0.495	0.085	0.808	0.057	0.605	0.041
RandomForest	0.489	0.080	0.814	0.052	0.612	0.037
SVR	0.491	0.069	0.813	0.053	0.613	0.037

Selected candidate internal output

Validation	0.416	0.819	0.632
Scaffold-held-out test	0.541	0.759	0.580

External-source validation output

Source	BindingDB REST
Exact training-structure overlaps removed	1091
External non-overlap n	160
External R2 / RMSE / MAE / Spearman	0.435 / 0.829 / 0.650 / 0.653
accession matches	True
exact structure overlap removed	True
minimum 30 external compounds	True
relation and assay context documented	False
Overall strict quality-gates passed	False
External status	external_source_exploratory_only

3.12. KIT - CHEMBL1936

Protein: Mast/stem cell growth factor receptor Kit | **UniProt:** P10721 | **Model version:** baseline_v1 | **Selected model:** ExtraTrees

Dataset and split output

Curated compound count	1946
Scaffold count	847
Train / validation / scaffold test	1350 / 248 / 348
Train / validation / test scaffolds	541 / 136 / 170
Fingerprint	Morgan, radius 2, 2,048 bits
Candidate model SHA256	92f96a4831c7f55573c5a70ed472a20f74a3126bbd042bda70c070810ae24679

Cross-validation comparison output

MedianBaseline	-0.038	0.055	0.913	0.023	0.728	0.007
ExtraTrees	0.320	0.057	0.739	0.033	0.581	0.020
RandomForest	0.312	0.062	0.743	0.037	0.586	0.019
SVR	0.294	0.028	0.753	0.021	0.593	0.015

Selected candidate internal output

Validation	0.337	0.878	0.664
Scaffold-held-out test	0.420	0.750	0.570

External-source validation output

Source	BindingDB REST
Exact training-structure overlaps removed	306
External non-overlap n	128
External R2 / RMSE / MAE / Spearman	-0.203 / 1.010 / 0.798 / -0.175
accession matches	True
exact structure overlap removed	True
minimum 30 external compounds	True
relation and assay context documented	False
Overall strict quality-gates passed	False
External status	external_source_exploratory_only

3.13. MAP2K1 - CHEMBL3587

Protein: Dual specificity mitogen-activated protein kinase kinase 1 | **UniProt:** Q02750 | **Model version:** baseline_v1 | **Selected model:** SVR

Dataset and split output

Curated compound count	866
Scaffold count	308
Train / validation / scaffold test	529 / 125 / 212
Train / validation / test scaffolds	196 / 50 / 62
Fingerprint	Morgan, radius 2, 2,048 bits
Candidate model SHA256	c6bf975e6857e9cbe3d9b4e3c721aac1abc48c46543343878756c349e52c5f86

Cross-validation comparison output

MedianBaseline	-0.033	0.047	1.001	0.081	0.801	0.035
ExtraTrees	0.250	0.114	0.850	0.057	0.640	0.036
RandomForest	0.283	0.102	0.831	0.046	0.631	0.028
SVR	0.277	0.181	0.830	0.075	0.643	0.074

Selected candidate internal output

Validation	0.417	0.749	0.597
Scaffold-held-out test	0.419	0.767	0.613

External-source validation output

Source	BindingDB REST
Exact training-structure overlaps removed	99
External non-overlap n	63
External R2 / RMSE / MAE / Spearman	0.447 / 0.751 / 0.627 / 0.606
accession matches	True
exact structure overlap removed	True
minimum 30 external compounds	True
relation and assay context documented	False
Overall strict quality-gates passed	False
External status	external_source_exploratory_only

3.14. MAPK14 - CHEMBL260

Protein: Mitogen-activated protein kinase 14 | **UniProt:** Q16539 | **Model version:** baseline_v1 | **Selected model:** SVR

Dataset and split output

Curated compound count	4744
Scaffold count	1717
Train / validation / scaffold test	3070 / 635 / 1039
Train / validation / test scaffolds	1098 / 275 / 344
Fingerprint	Morgan, radius 2, 2,048 bits
Candidate model SHA256	3da56275f01306bacb4de039023a95f15c0a5d0485fb7a5048a34500fdd91f12

Cross-validation comparison output

MedianBaseline	-0.011	0.005	1.102	0.056	0.896	0.044
ExtraTrees	0.509	0.035	0.767	0.023	0.590	0.018
RandomForest	0.502	0.040	0.772	0.029	0.599	0.022
SVR	0.528	0.031	0.752	0.022	0.584	0.016

Selected candidate internal output

Validation	0.527	0.730	0.560
Scaffold-held-out test	0.546	0.720	0.550

External-source validation output

Source	BindingDB REST
Exact training-structure overlaps removed	525
External non-overlap n	245
External R2 / RMSE / MAE / Spearman	0.033 / 1.148 / 0.858 / 0.435
accession matches	True
exact structure overlap removed	True
minimum 30 external compounds	True
relation and assay context documented	False
Overall strict quality-gates passed	False
External status	external_source_exploratory_only

3.15. MET - CHEMBL3717

Protein: Hepatocyte growth factor receptor | **UniProt:** P08581 | **Model version:** baseline_v1 | **Selected model:** SVR

Dataset and split output

Curated compound count	4193
Scaffold count	1714
Train / validation / scaffold test	2656 / 753 / 784
Train / validation / test scaffolds	1096 / 275 / 343
Fingerprint	Morgan, radius 2, 2,048 bits
Candidate model SHA256	a15e75b06d1877d95d3eda365289af214c214a50517828599503494d629589c1

Cross-validation comparison output

MedianBaseline	-0.012	0.007	1.105	0.047	0.867	0.053
ExtraTrees	0.484	0.009	0.789	0.030	0.596	0.028
RandomForest	0.465	0.009	0.804	0.033	0.608	0.031
SVR	0.507	0.060	0.770	0.046	0.593	0.031

Selected candidate internal output

Validation	0.560	0.679	0.505
Scaffold-held-out test	0.629	0.647	0.490

External-source validation output

Source	BindingDB REST
Exact training-structure overlaps removed	576
External non-overlap n	83
External R2 / RMSE / MAE / Spearman	0.596 / 0.813 / 0.494 / 0.820
accession matches	True
exact structure overlap removed	True
minimum 30 external compounds	True
relation and assay context documented	False
Overall strict quality-gates passed	False
External status	external_source_exploratory_only

3.16. MTOR - CHEMBL2842

Protein: Serine/threonine-protein kinase mTOR | **UniProt:** P42345 | **Model version:** baseline_v1 | **Selected model:** SVR

Dataset and split output

Curated compound count	4354
Scaffold count	1728
Train / validation / scaffold test	2854 / 714 / 786
Train / validation / test scaffolds	1105 / 277 / 346
Fingerprint	Morgan, radius 2, 2,048 bits
Candidate model SHA256	a049a85ee3dad0fbf0e6ebcd307e9d17bfbe2d21a83e46cbc73a7919bd42695e

Cross-validation comparison output

MedianBaseline	-0.020	0.012	1.205	0.077	0.973	0.077
ExtraTrees	0.644	0.056	0.708	0.035	0.534	0.034
RandomForest	0.644	0.059	0.708	0.038	0.540	0.035
SVR	0.674	0.045	0.678	0.032	0.515	0.029

Selected candidate internal output

Validation	0.719	0.638	0.476
Scaffold-held-out test	0.747	0.603	0.457

External-source validation output

Source	BindingDB REST
Exact training-structure overlaps removed	540
External non-overlap n	28
External R2 / RMSE / MAE / Spearman	0.614 / 0.707 / 0.582 / 0.722
accession matches	True
exact structure overlap removed	True
minimum 30 external compounds	False
relation and assay context documented	False
Overall strict quality-gates passed	False
External status	external_source_exploratory_only

3.17. PDGFRA - CHEMBL2007

Protein: Platelet-derived growth factor receptor alpha | **UniProt:** P16234 | **Model version:** baseline_v1 | **Selected model:** SVR

Dataset and split output

Curated compound count	632
Scaffold count	362
Train / validation / scaffold test	447 / 81 / 104
Train / validation / test scaffolds	231 / 58 / 73
Fingerprint	Morgan, radius 2, 2,048 bits
Candidate model SHA256	85a2ee6bdcf24feafa6f5f80a993d91b6a4b08e5abc5573387446ef4b4d723a7

Cross-validation comparison output

MedianBaseline	-0.052	0.061	1.056	0.094	0.857	0.131
ExtraTrees	0.116	0.270	0.955	0.085	0.756	0.085
RandomForest	0.231	0.102	0.900	0.053	0.710	0.041
SVR	0.237	0.074	0.898	0.078	0.716	0.063

Selected candidate internal output

Validation	0.425	0.824	0.660
Scaffold-held-out test	0.409	0.798	0.615

External-source validation output

Source	BindingDB REST
Exact training-structure overlaps removed	82
External non-overlap n	71
External R2 / RMSE / MAE / Spearman	-1.072 / 1.073 / 0.884 / -0.084
accession matches	True
exact structure overlap removed	True
minimum 30 external compounds	True
relation and assay context documented	False
Overall strict quality-gates passed	False
External status	external_source_exploratory_only

3.18. PIK3CA - CHEMBL4005

Protein: Phosphatidylinositol 4,5-bisphosphate 3-kinase catalytic subunit alpha isoform | **UniProt:** P42336 | **Model version:** baseline_v1 | **Selected model:** SVR

Dataset and split output

Curated compound count	6465
Scaffold count	2694
Train / validation / scaffold test	4067 / 1119 / 1279
Train / validation / test scaffolds	1724 / 431 / 539
Fingerprint	Morgan, radius 2, 2,048 bits
Candidate model SHA256	468cd00be07c79da4f746279019a5764f83106c58cfe1a42096eea6407827960

Cross-validation comparison output

MedianBaseline	-0.006	0.009	1.197	0.087	0.951	0.083
ExtraTrees	0.627	0.033	0.727	0.025	0.534	0.026
RandomForest	0.607	0.025	0.747	0.035	0.549	0.030
SVR	0.662	0.027	0.692	0.027	0.520	0.028

Selected candidate internal output

Validation	0.680	0.642	0.483
Scaffold-held-out test	0.714	0.616	0.467

External-source validation output

Source	BindingDB REST
Exact training-structure overlaps removed	781
External non-overlap n	566
External R2 / RMSE / MAE / Spearman	0.149 / 1.202 / 0.958 / 0.396
accession matches	True
exact structure overlap removed	True
minimum 30 external compounds	True
relation and assay context documented	False
Overall strict quality-gates passed	False
External status	external_source_exploratory_only

3.19. RET - CHEMBL2041

Protein: Proto-oncogene tyrosine-protein kinase receptor Ret | **UniProt:** P07949 | **Model version:** baseline_v1 | **Selected model:** SVR

Dataset and split output

Curated compound count	3980
Scaffold count	1523
Train / validation / scaffold test	2576 / 535 / 869
Train / validation / test scaffolds	974 / 244 / 305
Fingerprint	Morgan, radius 2, 2,048 bits
Candidate model SHA256	4e2a275cea01590aa02e6889ac94b7e2c54cc8ca504301dffddcec8cd00573a4

Cross-validation comparison output

MedianBaseline	-0.006	0.006	0.960	0.068	0.719	0.047
ExtraTrees	0.498	0.058	0.678	0.068	0.497	0.048
RandomForest	0.504	0.056	0.674	0.073	0.495	0.055
SVR	0.523	0.066	0.661	0.083	0.482	0.059

Selected candidate internal output

Validation	0.720	0.590	0.440
Scaffold-held-out test	0.616	0.612	0.443

External-source validation output

Source	BindingDB REST
Exact training-structure overlaps removed	1487
External non-overlap n	95
External R2 / RMSE / MAE / Spearman	-1.207 / 1.413 / 1.151 / 0.278
accession matches	True
exact structure overlap removed	True
minimum 30 external compounds	True
relation and assay context documented	False
Overall strict quality-gates passed	False
External status	external_source_exploratory_only

3.20. SRC - CHEMBL267

Protein: Proto-oncogene tyrosine-protein kinase Src | **UniProt:** P12931 | **Model version:** baseline_v1 | **Selected model:** ExtraTrees

Dataset and split output

Curated compound count	3601
Scaffold count	1546
Train / validation / scaffold test	2402 / 480 / 719
Train / validation / test scaffolds	988 / 248 / 310
Fingerprint	Morgan, radius 2, 2,048 bits
Candidate model SHA256	450aa84e9a126478a383b7f14afb1849748aaacdbb8b8802d5733b4c230ffde5

Cross-validation comparison output

MedianBaseline	-0.008	0.006	1.330	0.101	1.106	0.097
ExtraTrees	0.643	0.070	0.785	0.019	0.590	0.025
RandomForest	0.631	0.068	0.799	0.015	0.606	0.024
SVR	0.632	0.085	0.796	0.035	0.605	0.040

Selected candidate internal output

Validation	0.654	0.743	0.551
Scaffold-held-out test	0.733	0.841	0.610

External-source validation output

Source	BindingDB REST
Exact training-structure overlaps removed	403
External non-overlap n	62
External R2 / RMSE / MAE / Spearman	0.731 / 0.638 / 0.485 / 0.888
accession matches	True
exact structure overlap removed	True
minimum 30 external compounds	True
relation and assay context documented	False
Overall strict quality-gates passed	False
External status	external_source_exploratory_only

4. Strict validation gate definition

Target accession	External target accession equals local target UniProt accession.
Endpoint	All retained records have standard_type IC50.
Unit	All retained records are normalized to nM.
Relation	All retained records have exact relation =.
Assay context	Every retained record has documented assay context.
Publication identifier	Every retained record has a documented publication ID.
Structure independence	No canonical external structure remains in training data.
Publication independence	No publication ID overlap remains with training provenance.
Minimum sample size	At least 30 external non-overlap compounds remain.
Metric floor	External RMSE <= 1.0; MAE <= 0.75; Spearman >= 0.50.
Reproducibility	Saved model SHA256 matches report candidate hash.
Source declared	Source name and source URL are non-empty.

5. Application outputs and local artifacts

Single prediction output: target ID/gene, input and canonical SMILES, predicted pIC50, predicted IC50 nM, dataset-level typical error, nearest training Tanimoto similarity, confidence label, model version and algorithm. Batch CSV output additionally provides a row-level status and error field. All outputs can be downloaded from the application as CSV. Prediction history is stored only in the browser local storage and can be exported as CSV.

Local target artifact layout

data/targets/CHEMBL*/target_metadata.json, protein.fasta, IC50_raw.csv, IC50_clean.csv; models/CHEMBL*/baseline_v*/candidate_model.joblib, evaluation_report.json, model_comparison_cv.csv, split_assignments.csv, external_validation_report.json, external_validation_predictions.csv; models/target_registry.json; data/external_validation/pubchem_assay_inventory.json; data/external_validation/pubchem_independent/.

Final status: The tool is technically ready for target-specific screening across 20 registered targets. External validation remains an active evidence-collection and qualification task. Do not call the present models commercial-grade or clinically/experimentally validated until every required gate has passed and an expert review has been completed.